The goal of Week 3 is to improve Linux server security by addressing unresolved issues from Weeks 1 and 2 and measuring progress using Lynis before-and-after security scores.

Key tasks:

1. **Remove unnecessary services:** Check for unused Ubuntu services and remove the `telnetd` package if it is not needed.
2. **Secure `legacy-report-tool`:** Restrict network access, fix world-writable configuration permissions, run it with a less-privileged account, and apply AppArmor confinement.
3. **Strengthen AppArmor**: Change profiles from complain mode to enforce mode and activate inactive profiles where appropriate.
4. **Audit file permissions**: Identify and correct insecure permissions beyond `config.yml`.
5. **Harden the kernel:** Apply appropriate security parameters for ASLR and network-level protections.
6. **Measure improvements**: Run Lynis before and after the changes and compare the security scores.

Step 1. Run a Lynis baseline before touching anything



```
newadmin@ops-01:~$ sudo apt install lynis -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  dnsutils apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu | kde-runtime | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 48 not upgraded.
Need to get 581 kB of archives.
After this operation, 3164 kB of additional disk space will be used.
Ign:1 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 lynis all 3.0.7-1
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 menu amd64 2.1.47ubuntu4
Ign:1 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 lynis all 3.0.7-1
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 menu amd64 2.1.47ubuntu4
Ign:1 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 lynis all 3.0.7-1
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 menu amd64 2.1.47ubuntu4
Err:1 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 lynis all 3.0.7-1
  Temporary failure resolving 'gb.archive.ubuntu.com'
Err:2 http://gb.archive.ubuntu.com/ubuntu jammy/universe amd64 menu amd64 2.1.47ubuntu4
  Temporary failure resolving 'gb.archive.ubuntu.com'
E: Failed to fetch http://gb.archive.ubuntu.com/ubuntu/pool/universe/l/lynis/lynis_3.0.7-1_all.deb  Temporary failure re
solving 'gb.archive.ubuntu.com'
E: Failed to fetch http://gb.archive.ubuntu.com/ubuntu/pool/universe/m/menu/menu_2.1.47ubuntu4_amd64.deb  Temporary fail
ure resolving 'gb.archive.ubuntu.com'
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
newadmin@ops-01:~$ sudo lynis audit system --cronjob | tee ~/lynis-before.txt
sudo: lynis: command not found
newadmin@ops-01:~$ grep -E "Hardening index|Warning|Suggestion" ~/lynis-before.txt | head -30
```

Step 2. Remove unnecessary services and packages

Step 2. Remove unnecessary services and packages

Week 3 Lab · System Hardening · Step 2 of 8

Ubuntu minimal installs are leaner than desktop installs, but may still include services that have no place on a hardened server.

```
# List all running services - read the output carefully before disabling an
systemctl list-units --type=service --state=running

# Check what is actually listening
sudo ss -tulpn

# Remove the telnet packages
sudo apt purge -y inetutils-telnetd openbsd-inetd
sudo apt autoremove -y

# Review other candidates
sudo systemctl status multipathd
sudo systemctl status rpcbind
```

Before you disable or mask anything, find out what it does and what depends on it:

```
systemctl cat <service>
systemctl list-dependencies --reverse <service>
```

For a service you decide to stop for good (shown here for multipathd):

```
sudo systemctl disable --now multipathd
sudo systemctl mask multipathd
# Check that multipathd service is inactive:
sudo systemctl status multipathd
```

- [ ] I reviewed each candidate before changing it.
- [ ] I saved sudo ss -tulpn after the removal (evidence item 3).

› Tick every box above to complete this step.

PROMPTS ›› services

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

PROMPT 1

Look closely at your `sudo ss -tulpn` output. Why did rpcbind/telnet show up there, but multipathd did not?

→hint

PROMPT 2

If multipathd doesn't open a network port, why do we still disable and mask it during a hardening audit?

→hint

PROMPT 3

Why remove the telnet packages instead of only stopping the service?

→hint

0/3 hints opened

```
newadmin@ops-01:~$ systemctl list-units --type=service --state=running
  UNIT                          LOAD   ACTIVE SUB     DESCR
IPTION
  dbus.service                  loaded active running D-Bus System Message Bus
  getty@tty1.service            loaded active running Getty on tty1
  inetd.service                 loaded active running Internet superserver
  inetutils-syslogd.service     loaded active running LSB: GNU Network Utilities syslog daemon
  legacy-report.service         loaded active running Legacy Report Tool
  mariadb.service               loaded active running MariaDB 10.6.23 database server
  multipathd.service            loaded active running Device-Mapper Multipath Device Controller
  networkd-dispatcher.service loaded active running Dispatcher daemon for systemd-networkd
  ssh.service                   loaded active running OpenBSD Secure Shell server
  systemd-journald.service      loaded active running Journal Service
  systemd-logind.service        loaded active running User Login Management
  systemd-networkd.service      loaded active running Network Configuration
  systemd-resolved.service      loaded active running Network Name Resolution
  systemd-timesyncd.service     loaded active running Network Time Synchronization
  systemd-udevd.service         loaded active running Rule-based Manager for Device Events and Files
  unattended-upgrades.service loaded active running Unattended Upgrades Shutdown
  user@1000.service             loaded active running User Manager for UID 1000
```

```
newadmin@ops-01:~$ sudo ss -tulpn
Netid State  Recv-Q  Send-Q           Local Address:Port   Peer Address:Port Process
udp   UNCONN 0       0                127.0.0.53%lo:53          0.0.0.0:*     users:(("systemd-resolve",pid=596,fd=13))
udp   UNCONN 0       0        192.168.56.101%enp0s8:68          0.0.0.0:*     users:(("systemd-network",pid=594,fd=15))
tcp   LISTEN 0       128                    0.0.0.0:22          0.0.0.0:*     users:(("sshd",pid=1262,fd=3))
tcp   LISTEN 0       128                    0.0.0.0:23          0.0.0.0:*     users:(("inetd",pid=614,fd=7))
tcp   LISTEN 0       80                 127.0.0.1:3306        0.0.0.0:*     users:(("mariadbd",pid=694,fd=21))
tcp   LISTEN 0       4096             127.0.0.53%lo:53          0.0.0.0:*     users:(("systemd-resolve",pid=596,fd=14))
tcp   LISTEN 0       5                      0.0.0.0:8080        0.0.0.0:*     users:(("python3",pid=618,fd=3))
tcp   LISTEN 0       128                       [::]:22             [::]:*     users:(("sshd",pid=1262,fd=4))
newadmin@ops-01:~$ sudo apt purge -y inetutils-telnetd openbsd-inetd
sudo apt autoremove -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  inetutils-syslogd libevent-2.1-7 tcpd update-inetd
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  inetutils-telnetd* openbsd-inetd*
0 upgraded, 0 newly installed, 2 to remove and 48 not upgraded.
After this operation, 265 kB disk space will be freed.
(Reading database ... 66055 files and directories currently installed.)
Removing inetutils-telnetd (2:2.2-2ubuntu0.2) ...
Removing openbsd-inetd (0.20160825-5) ...
(Reading database ... 66038 files and directories currently installed.)
Purging configuration files for inetutils-telnetd (2:2.2-2ubuntu0.2) ...
Purging configuration files for openbsd-inetd (0.20160825-5) ...##################.............................]
Progress: [ 67%] [##########################################################################.....................]

Progress: [ 78%] [#################################################################################.....................]
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```

Activate Windows

```
newadmin@ops-01:~$ systemctl cat <service>
systemctl list-dependencies --reverse <service>
-bash: syntax error near unexpected token `newline'
-bash: syntax error near unexpected token `newline'
newadmin@ops-01:~$ sudo systemctl disable --now multipathd
sudo systemctl mask multipathd
# Check that multipathd service is inactive:
sudo systemctl status multipathd
Removed /etc/systemd/system/sysinit.target.wants/multipathd.service.
Removed /etc/systemd/system/multipath-tools.service.
Removed /etc/systemd/system/sockets.target.wants/multipathd.socket.
Warning: Stopping multipathd.service, but it can still be activated by:
  multipathd.socket
Created symlink /etc/systemd/system/multipathd.service → /dev/null.
○ multipathd.service
     Loaded: masked (Reason: Unit multipathd.service is masked.)
     Active: inactive (dead) since Tue 2026-10-13 09:08:49 UTC; 242ms ago
TriggeredBy: ● multipathd.socket
   Main PID: 426 (code=exited, status=0/SUCCESS)
     Status: "shutdown"
        CPU: 834ms

Oct 13 08:15:45 ops-01 multipathd[426]: --------start up--------
Oct 13 08:15:45 ops-01 multipathd[426]: read /etc/multipath.conf
Oct 13 08:15:45 ops-01 multipathd[426]: path checkers start up
Oct 13 08:15:45 ops-01 systemd[1]: Started Device-Mapper Multipath Device Controller.
Oct 13 09:08:49 ops-01 multipathd[426]: exit (signal)
Oct 13 09:08:49 ops-01 multipathd[426]: --------shut down-------
Oct 13 09:08:49 ops-01 systemd[1]: Stopping Device-Mapper Multipath Device Controller...
Oct 13 09:08:49 ops-01 systemd[1]: multipathd.service: Deactivated successfully.
Oct 13 09:08:49 ops-01 systemd[1]: Stopped Device-Mapper Multipath Device Controller.
newadmin@ops-01:~$
```

Activate Windows
Go to Settings to activate Windows.

Step 2. Remove unnecessary services and packages

Week 3 Lab · System Hardening · Step 2 of 8

Ubuntu minimal installs are leaner than desktop installs, but may still include services that have no place on a hardened server.

```
# List all running services - read the output carefully before disabling anything
systemctl list-units --type=service --state=running

# Check what is actually listening
sudo ss -tulpn

# Remove the telnet packages
sudo apt purge -y inetutils-telnetd openbsd-inetd
sudo apt autoremove -y

# Review other candidates
sudo systemctl status multipathd
sudo systemctl status rpcbind
```

Before you disable or mask anything, find out what it does and what depends on it:

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systemctl cat <service>
systemctl list-dependencies --reverse <service>
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For a service you decide to stop for good (shown here for multipathd):

```
sudo systemctl disable --now multipathd
sudo systemctl mask multipathd
# Check that multipathd service is inactive:
sudo systemctl status multipathd
```

- [x] I reviewed each candidate before changing it.
- [x] I saved `sudo ss -tulpn` after the removal (evidence item 3).

✓ Slide complete

PRACTICE ›› services

What is the difference between `systemctl disable` and `systemctl mask`?

A `disable` stops the service now; `mask` only stops it starting at boot.

B `disable` removes automatic start but the service can still be started manually or by another service; `mask` makes it unstartable until it is unmasked.

C `mask` uninstalls the package; `disable` only stops the running process.

PROMPTS ›› services

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

HINT

Think about what kind of resource a service needs to open to appear in an 'ss' network socket audit.

LOOK AT Your terminal output from `sudo ss -tulpn` VERSUS `systemctl status multipathd`.

LOG Explain that network-facing services open network ports (TCP/UDP) and show up in ss, while local daemons like multipathd run background system tasks without opening network sockets.

HINT

Consider the overall attack surface. Does a vulnerability have to come from the network to be dangerous to a local system?

LOOK AT Research local privilege escalation (LPE) risks and system resource management.

LOG State how reducing the attack surface means removing unused local code that could be exploited locally (LPE), while also reclaiming memory and CPU cycles.

HINT

Think about what the installed package still provides once the process is stopped, and what could start it again.

LOOK AT `dpkg -l | grep -i telnet; systemctl is-enabled`.

LOG State what risk stopping alone would have left in place.

3/3 hints opened

Activate Windows
Go to Settings to activate Windows.

Step 3. Fix the legacy-report-tool properly

```
newadmin@ops-01:~$ sudo journalctl -u legacy-report.service --since "1 week ago" | grep -c "GET /"
0
newadmin@ops-01:~$ sudo systemctl list-dependencies legacy-report.service --reverse
legacy-report.service
● └─multi-user.target
●   └─graphical.target
newadmin@ops-01:~$
```

```
 GNU nano 6.2                                /opt/legacy-report-tool/app.py
!/usr/bin/env python3
 Originally deployed by a.chen — 2024-02-20
 TODO: add /status auth before go-live — never happened
 Originally deployed by a.chen — 2024-02-20
 TODO: add /status auth before go-live — never happened
mport http.server, yaml, json

ONFIG = yaml.safe_load(open('/opt/legacy-report-tool/config.yml'))

lass Handler(http.server.BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header('Content-type', 'application/json')
        self.end_headers()
        if self.path == '/status':
            self.wfile.write(json.dumps(CONFIG).encode())
        else:
            self.wfile.write(b'{"status":"running","app":"legacy-report-tool","version":"1.2.3"}')
    def log_message(self, format, *args):
        pass

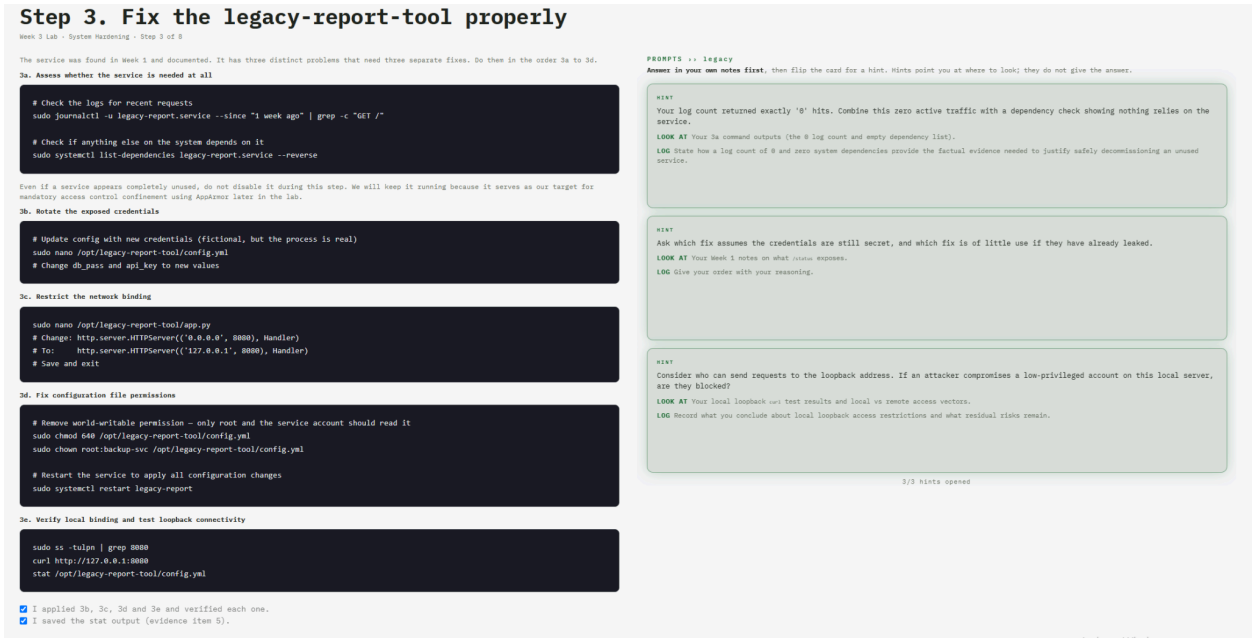
ttp.server.HTTPServer(('0.0.0.0', 8080), Handler).serve_forever()


                            [ Read 22 lines ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

```
○ multipathd.service
     Loaded: masked (Reason: Unit multipathd.service is masked.)
     Active: inactive (dead) since Tue 2026-10-13 09:08:49 UTC; 242ms ago
TriggeredBy: ● multipathd.socket
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Oct 13 09:08:49 ops-01 systemd[1]: Stopping Device-Mapper Multipath Device Controller...
Oct 13 09:08:49 ops-01 systemd[1]: multipathd.service: Deactivated successfully.
Oct 13 09:08:49 ops-01 systemd[1]: Stopped Device-Mapper Multipath Device Controller.
newadmin@ops-01:~$ sudo journalctl -u legacy-report.service --since "1 week ago" | grep -c "GET /"
0
newadmin@ops-01:~$ sudo systemctl list-dependencies legacy-report.service --reverse
legacy-report.service
● └─multi-user.target
●   └─graphical.target
newadmin@ops-01:~$ sudo nano /opt/legacy-report-tool/config.yml
newadmin@ops-01:~$
newadmin@ops-01:~$ sudo nano /opt/legacy-report-tool/app.py
newadmin@ops-01:~$
newadmin@ops-01:~$
```

```
ewadmin@ops-01:~$ sudo ss -tulpn | grep 8080
url http://127.0.0.1:8080
tat /opt/legacy-report-tool/config.yml
cp   LISTEN 0      5                    0.0.0.0:8080      0.0.0.0:*    users:(("python3",pid=618,fd=3))
"status":"running","app":"legacy-report-tool","version":"1.2.3"}  File: /opt/legacy-report-tool/config.yml
 Size: 101             Blocks: 8          IO Block: 4096   regular file
evice: fd00h/64768d    Inode: 393281      Links: 1
ccess: (0640/-rw-r-----)  Uid: (    0/    root)   Gid: ( 1001/backup-svc)
ccess: 2026-10-13 08:15:49.139999960 +0000
odify: 2026-10-13 09:12:16.671979593 +0000
hange: 2026-10-13 09:22:51.743975775 +0000
Birth: 2026-09-22 08:17:33.534055972 +0000
```



Step 3. Fix the legacy-report-tool properly

Week 3 Lab · System Hardening · Step 3 of 5

The service was found in Week 1 and documented. It has three distinct problems that need three separate fixes. Do them in the order 3a to 3d.

3a. Assess whether the service is needed at all

```
# Check the logs for recent requests
sudo journalctl -u legacy-report.service --since "1 week ago" | grep -c "GET /"

# Check if anything else on the system depends on it
sudo systemctl list-dependencies legacy-report.service --reverse
```

Even if a service appears completely unused, do not disable it during this step. We will keep it running because it serves as our target for mandatory access control confinement using AppArmor later in the lab.

3b. Rotate the exposed credentials

```
# Update config with new credentials (fictional, but the process is real)
sudo nano /opt/legacy-report-tool/config.yml
# Change db_pass and api_key to new values
```

3c. Restrict the network binding

```
sudo nano /opt/legacy-report-tool/app.py
# Change: http.server.HTTPServer(('0.0.0.0', 8080), Handler)
# To:     http.server.HTTPServer(('127.0.0.1', 8080), Handler)
# Save and exit
```

3d. Fix configuration file permissions

```
# Remove world-writable permission — only root and the service account should read it
sudo chmod 640 /opt/legacy-report-tool/config.yml
sudo chown root:backup-svc /opt/legacy-report-tool/config.yml

# Restart the service to apply all configuration changes
sudo systemctl restart legacy-report
```

3e. Verify local binding and test loopback connectivity

```
sudo ss -tulpn | grep 8080
curl http://127.0.0.1:8080
stat /opt/legacy-report-tool/config.yml
```

- [x] I applied 3b, 3c, 3d and 3e and verified each one.
- [x] I saved the stat output (evidence item 5).

PROMPTS ›› legacy

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

HINT

Your log count returned exactly '0' hits. Combine this zero active traffic with a dependency check showing nothing relies on the service.

LOOK AT Your 3a command outputs (the 0 log count and empty dependency list).

LOG State how a log count of 0 and zero system dependencies provide the factual evidence needed to justify safely decommissioning an unused service.

HINT

Ask which fix assumes the credentials are still secret, and which fix is of little use if they have already leaked.

LOOK AT Your Week 1 notes on what `/status` exposes.

LOG Give your order with your reasoning.

HINT

Consider who can send requests to the loopback address. If an attacker compromises a low-privileged account on this local server, are they blocked?

LOOK AT Your local loopback `curl` test results and local vs remote access vectors.

LOG Record what you conclude about local loopback access restrictions and what residual risks remain.

3/3 hints opened

Step 4. Audit system-wide file permissions

```
newadmin@ops-01:~$ sudo find / -xdev -type f -perm -o+w 2>/dev/null
newadmin@ops-01:~$ sudo find / -xdev \( -perm -4000 -o -perm -2000 \) -type f 2>/dev/null
/usr/bin/fusermount3
/usr/bin/sudo
/usr/bin/chage
/usr/bin/ssh-agent
/usr/bin/pkexec
/usr/bin/chsh
/usr/bin/umount
/usr/bin/newgrp
/usr/bin/su
/usr/bin/expiry
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/mount
/usr/bin/gpasswd
/usr/libexec/polkit-agent-helper-1
/usr/sbin/unix_chkpwd
/usr/sbin/pam_extrausers_chkpwd
/usr/lib/openssh/ssh-keysign
/usr/lib/mysql/plugin/auth_pam_tool_dir/auth_pam_tool
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
newadmin@ops-01:~$ ls -la /tmp/
total 44
drwxrwxrwt 11 root root 4096 Oct 13 09:07 .
drwxr-xr-x 20 root root 4096 Sep 21 11:09 ..
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .ICE-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .Test-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .X11-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .XIM-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .font-unix
drwx------  2 root root 4096 Oct 13 08:15 snap-private-tmp
drwx------  3 root root 4096 Oct 13 08:15 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-logind.service-N73C4F
drwx------  3 root root 4096 Oct 13 08:15 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-resolved.service-khVh9a
drwx------  3 root root 4096 Oct 13 09:07 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-timesyncd.service-huxnL9
newadmin@ops-01:~$ sudo find / -xdev -type f -perm -o+w 2>/dev/null
newadmin@ops-01:~$ sudo find / -xdev \( -perm -4000 -o -perm -2000 \) -type f 2>/dev/null
/usr/bin/fusermount3
/usr/bin/sudo
/usr/bin/chage
/usr/bin/ssh-agent
/usr/bin/pkexec
/usr/bin/chsh
/usr/bin/umount
/usr/bin/newgrp
/usr/bin/su
```

```
newadmin@ops-01:~$ sudo find / -xdev -type f -perm -o+w 2>/dev/null
newadmin@ops-01:~$ sudo find / -xdev \( -perm -4000 -o -perm -2000 \) -type f 2>/dev/null
/usr/bin/fusermount3
/usr/bin/sudo
/usr/bin/chage
/usr/bin/ssh-agent
/usr/bin/pkexec
/usr/bin/chsh
/usr/bin/umount
/usr/bin/newgrp
/usr/bin/su
/usr/bin/expiry
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/mount
/usr/bin/gpasswd
/usr/libexec/polkit-agent-helper-1
/usr/sbin/unix_chkpwd
/usr/sbin/pam_extrausers_chkpwd
/usr/lib/openssh/ssh-keysign
/usr/lib/mysql/plugin/auth_pam_tool_dir/auth_pam_tool
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
newadmin@ops-01:~$ ls -la /tmp/
total 44
drwxrwxrwt 11 root root 4096 Oct 13 09:07 .
drwxr-xr-x 20 root root 4096 Sep 21 11:09 ..
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .ICE-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .Test-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .X11-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .XIM-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .font-unix
drwx------  2 root root 4096 Oct 13 08:15 snap-private-tmp
drwx------  3 root root 4096 Oct 13 08:15 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-logind.service-N73C4F
drwx------  3 root root 4096 Oct 13 08:15 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-resolved.service-khVh9a
drwx------  3 root root 4096 Oct 13 09:07 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-timesyncd.service-huxnL9
newadmin@ops-01:~$
```

Step 4. Audit system-wide file permissions

Week 3 Lab · System Hardening · Step 4 of 8

You already secured the legacy tool configuration file. Now, run a broader system-wide scan to verify that no other unexpected files or binaries have dangerous permissions.

```
# Find world-writable files on the primary filesystem
sudo find / -xdev -type f -perm -o+w 2>/dev/null

# Audit SUID/SGID binaries on the primary filesystem
sudo find / -xdev \( -perm -4000 -o -perm -2000 \) -type f 2>/de

# Check for stray files or subdirectories in temporary storage
ls -la /tmp/
```

Review the output logs carefully. Document any unexpected findings that differ from a clean, hardened baseline system.

- [x] I ran all three checks and saved the output.
- [x] I recorded the mode before and after for every change, with my reason.

✓ Slide complete

PRACTICE ›› perms

A file owned by root has mode `755`. Who can do what?

A The owner can read, write and execute; group and others can read and execute.

B The owner can read and write; group and others can only read.

C Everyone can read, write and execute.

PROMPTS ›› perms

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

HINT

Decide what each file is for first: who needs to read it, who needs to write it, and does anything execute it?

LOOK AT Your `find` output and `stat` on each finding.

LOG For each finding, the mode before, the mode after, and why.

HINT

Compare each entry with what you would expect on a minimal server, and find out which package owns any you do not recognise.

LOOK AT `dpkg -S <path>`; online references for SUID binaries.

LOG List anything unexpected and what you did about it.

2/2 hints opened

Step 5. Kernel hardening with sysctl

```
newadmin@ops-01:~$ sudo find / -xdev \( -perm -4000 -o -perm -2000 \) -type f 2>/dev/null
/usr/bin/fusermount3
/usr/bin/sudo
/usr/bin/chage
/usr/bin/ssh-agent
/usr/bin/pkexec
/usr/bin/chsh
/usr/bin/umount
/usr/bin/newgrp
/usr/bin/su
/usr/bin/expiry
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/mount
/usr/bin/gpasswd
/usr/libexec/polkit-agent-helper-1
/usr/sbin/unix_chkpwd
/usr/sbin/pam_extrausers_chkpwd
/usr/lib/openssh/ssh-keysign
/usr/lib/mysql/plugin/auth_pam_tool_dir/auth_pam_tool
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
newadmin@ops-01:~$ ls -la /tmp/
total 44
drwxrwxrwt 11 root root 4096 Oct 13 09:07 .
drwxr-xr-x 20 root root 4096 Sep 21 11:09 ..
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .ICE-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .Test-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .X11-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .XIM-unix
drwxrwxrwt  2 root root 4096 Oct 13 08:15 .font-unix
drwx------  2 root root 4096 Oct 13 08:15 snap-private-tmp
drwx------  3 root root 4096 Oct 13 08:15 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-logind.service-N73C4F
drwx------  3 root root 4096 Oct 13 08:15 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-resolved.service-khVh9a
drwx------  3 root root 4096 Oct 13 09:07 systemd-private-0e87a3007cd545799b9cc7fd2c414d8c-systemd-timesyncd.service-huxnL9
newadmin@ops-01:~$ sudo nano /etc/sysctl.d/99-hardening.conf
newadmin@ops-01:~$ kernel.randomize_va_space = 2
fs.suid_dumpable = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.icmp_ignore_bogus_error_responses = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv6.conf.all.accept_source_route = 0
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
net.ipv6.conf.all.accept_redirects = 0
net.ipv4.conf.all.log_martians = 1
-bash: kernel.randomize_va_space: command not found
-bash: fs.suid_dumpable: command not found
-bash: net.ipv4.icmp_echo_ignore_broadcasts: command not found
-bash: net.ipv4.icmp_ignore_bogus_error_responses: command not found
-bash: net.ipv4.conf.all.accept_source_route: command not found
-bash: net.ipv6.conf.all.accept_source_route: command not found
-bash: net.ipv4.tcp_syncookies: command not found
-bash: net.ipv4.conf.all.accept_redirects: command not found
-bash: net.ipv4.conf.all.send_redirects: command not found
-bash: net.ipv6.conf.all.accept_redirects: command not found
```

Activate Window

```
newadmin@ops-01:~$ sudo sysctl --system
sysctl kernel.randomize_va_space
* Applying /etc/sysctl.d/10-console-messages.conf ...
kernel.printk = 4 4 1 7
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
kernel.kptr_restrict = 1
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
kernel.sysrq = 176
* Applying /etc/sysctl.d/10-network-security.conf ...
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
* Applying /etc/sysctl.d/10-ptrace.conf ...
kernel.yama.ptrace_scope = 1
* Applying /etc/sysctl.d/10-zeropage.conf ...
vm.mmap_min_addr = 65536
* Applying /usr/lib/sysctl.d/50-default.conf ...
kernel.core_uses_pid = 1
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.default.accept_source_route = 0
sysctl: setting key "net.ipv4.conf.all.accept_source_route": Invalid argument
net.ipv4.conf.default.promote_secondaries = 1
sysctl: setting key "net.ipv4.conf.all.promote_secondaries": Invalid argument
net.ipv4.ping_group_range = 0 2147483647
net.core.default_qdisc = fq_codel
fs.protected_hardlinks = 1
fs.protected_symlinks = 1
fs.protected_regular = 1
fs.protected_fifos = 1
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
kernel.pid_max = 4194304
* Applying /etc/sysctl.d/99-hardening.conf ...
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
fs.protected_fifos = 1
fs.protected_hardlinks = 1
fs.protected_regular = 2
fs.protected_symlinks = 1
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.conf ...
kernel.randomize_va_space = 2
newadmin@ops-01:~$
```

Step 5. Kernel hardening with sysctl

Week 3 Lab · System Hardening · Step 5 of 8

Linux kernel parameters can be set at runtime via `/proc/sys` and made persistent via `/etc/sysctl.d/`. Several security parameters are not enforced by default.

```
sudo nano /etc/sysctl.d/99-hardening.conf
```

Add these lines:

```
kernel.randomize_va_space = 2
fs.suid_dumpable = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.icmp_ignore_bogus_error_responses = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv6.conf.all.accept_source_route = 0
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
net.ipv6.conf.all.accept_redirects = 0
net.ipv4.conf.all.log_martians = 1
```

Apply without rebooting, then check a value:

```
sudo sysctl --system
sysctl kernel.randomize_va_space
```

- [x] I applied the settings and checked their values.
- [x] I wrote down what each parameter addresses in my own notes.

✓ Slide complete

PROMPTS ›› sysctl

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

HINT

Group them first: memory protections, ICMP handling, routing and redirects, SYN handling, logging. Then look each one up rather than guessing.

LOOK AT The kernel documentation for `ip-sysctl`; `sysctl(8)` on manpages.ubuntu.com.

LOG A short line per parameter, grouped.

HINT

Think about what an attacker needs to know about memory layout, and how they might learn it another way.

LOOK AT Search for ASLR and information-leak bypasses.

LOG One sentence on what it protects and one on its limits.

2/2 hints opened

Step 6. Activate AppArmor enforcement

```
newadmin@ops-01:~$ sudo apt update && sudo apt install -y apparmor-utils apparmor-profiles
Ign:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy InRelease
Ign:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates InRelease
Ign:4 http://gb.archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy InRelease
Ign:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates InRelease
Ign:4 http://gb.archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy InRelease
Ign:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates InRelease
Ign:4 http://gb.archive.ubuntu.com/ubuntu jammy-backports InRelease
Err:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:2 http://gb.archive.ubuntu.com/ubuntu jammy InRelease
  Temporary failure resolving 'gb.archive.ubuntu.com'
Err:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates InRelease
  Temporary failure resolving 'gb.archive.ubuntu.com'
Err:4 http://gb.archive.ubuntu.com/ubuntu jammy-backports InRelease
  Temporary failure resolving 'gb.archive.ubuntu.com'
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
48 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: Failed to fetch http://gb.archive.ubuntu.com/ubuntu/dists/jammy/InRelease  Temporary failure resolving 'gb.archive.ubuntu.com'
W: Failed to fetch http://gb.archive.ubuntu.com/ubuntu/dists/jammy-updates/InRelease  Temporary failure resolving 'gb.archive.ubuntu.com'
W: Failed to fetch http://gb.archive.ubuntu.com/ubuntu/dists/jammy-backports/InRelease  Temporary failure resolving 'gb.archive.ubuntu.com'
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/jammy-security/InRelease  Temporary failure resolving 'security.ubuntu.com'
W: Some index files failed to download. They have been ignored, or old ones used instead.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-apparmor python3-libapparmor
Suggested packages:
  vim-addon-manager
The following NEW packages will be installed:
  apparmor-profiles apparmor-utils python3-apparmor python3-libapparmor
0 upgraded, 4 newly installed, 0 to remove and 48 not upgraded.
Need to get 202 kB of archives.
After this operation, 1550 kB of additional disk space will be used.
Ign:1 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-libapparmor amd64 3.0.4-2ubuntu2.5
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-apparmor all 3.0.4-2ubuntu2.5
Ign:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apparmor-utils all 3.0.4-2ubuntu2.5
Ign:4 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apparmor-profiles all 3.0.4-2ubuntu2.5
Ign:1 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-libapparmor amd64 3.0.4-2ubuntu2.5
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-apparmor all 3.0.4-2ubuntu2.5
Ign:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apparmor-utils all 3.0.4-2ubuntu2.5
Ign:4 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apparmor-profiles all 3.0.4-2ubuntu2.5
Ign:1 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-libapparmor amd64 3.0.4-2ubuntu2.5
Ign:2 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-apparmor all 3.0.4-2ubuntu2.5
Ign:3 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apparmor-utils all 3.0.4-2ubuntu2.5
Ign:4 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apparmor-profiles all 3.0.4-2ubuntu2.5
Err:1 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-libapparmor amd64 3.0.4-2ubuntu2.5
  Temporary failure resolving 'gb.archive.ubuntu.com'
Err:2 http://gb.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-apparmor all 3.0.4-2ubuntu2.5
```

```
newadmin@ops-01:~$ sudo aa-status
apparmor module is loaded.
8 profiles are loaded.
8 profiles are in enforce mode.
   /usr/lib/NetworkManager/nm-dhcp-client.action
   /usr/lib/NetworkManager/nm-dhcp-helper
   /usr/lib/connman/scripts/dhclient-script
   /usr/lib/snapd/snap-confine
   /{,usr/}sbin/dhclient
   lsb_release
   nvidia_modprobe
   nvidia_modprobe//kmod
0 profiles are in complain mode.
0 profiles are in kill mode.
0 profiles are in unconfined mode.
0 processes have profiles defined.
0 processes are in enforce mode.
0 processes are in complain mode.
0 processes are unconfined but have a profile defined.
0 processes are in mixed mode.
0 processes are in kill mode.
newadmin@ops-01:~$
```

```
newadmin@ops-01:~$ sudo aa-status
apparmor module is loaded.
8 profiles are loaded.
8 profiles are in enforce mode.
   /usr/lib/NetworkManager/nm-dhcp-client.action
   /usr/lib/NetworkManager/nm-dhcp-helper
   /usr/lib/connman/scripts/dhclient-script
   /usr/lib/snapd/snap-confine
   /{,usr/}sbin/dhclient
   lsb_release
   nvidia_modprobe
   nvidia_modprobe//kmod
0 profiles are in complain mode.
0 profiles are in kill mode.
0 profiles are in unconfined mode.
0 processes have profiles defined.
0 processes are in enforce mode.
0 processes are in complain mode.
0 processes are unconfined but have a profile defined.
0 processes are in mixed mode.
0 processes are in kill mode.
newadmin@ops-01:~$ sudo aa-enforce /etc/apparmor.d/usr.sbin.avahi-daemon
sudo: aa-enforce: command not found
newadmin@ops-01:~$ ls /etc/apparmor.d/
abi  abstractions  disable  force-complain  local  lsb_release  nvidia_modprobe  sbin.dhclient  tunables  usr.lib.snapd.snap-confine.real  usr.sbin.mariadbd
newadmin@ops-01:~$ sudo aa-status
sudo aa-status > ~/week3-apparmor-status.txt
apparmor module is loaded.
8 profiles are loaded.
8 profiles are in enforce mode.
   /usr/lib/NetworkManager/nm-dhcp-client.action
   /usr/lib/NetworkManager/nm-dhcp-helper
   /usr/lib/connman/scripts/dhclient-script
   /usr/lib/snapd/snap-confine
   /{,usr/}sbin/dhclient
   lsb_release
   nvidia_modprobe
   nvidia_modprobe//kmod
0 profiles are in complain mode.
0 profiles are in kill mode.
0 profiles are in unconfined mode.
0 processes have profiles defined.
0 processes are in enforce mode.
0 processes are in complain mode.
0 processes are unconfined but have a profile defined.
0 processes are in mixed mode.
0 processes are in kill mode.
newadmin@ops-01:~$
```

Step 6. Activate AppArmor enforcement

Week 3 Lab · System Hardening · Step 6 of 8

AppArmor is installed on Ubuntu by default, but profiles may be in complain mode (violations are logged, not blocked) instead of enforce mode.

```
sudo apt update && sudo apt install -y apparmor-utils apparmor-

# Check current status of loaded profiles
sudo aa-status
```

Move an existing profile from complain mode into active enforcement mode to protect the local environment:

```
sudo aa-enforce /etc/apparmor.d/usr.sbin.avahi-daemon

# Review the profile directory structure
ls /etc/apparmor.d/
```

Verify that your profile enforcement changes are active in the kernel stack:

```
sudo aa-status
sudo aa-status > ~/week3-apparmor-status.txt
```

- [x] I confirmed that the required apparmor profiles exist before enforcing them.
- [x] I saved the aa-status output (evidence items 4 and 8).

✓ Slide complete

PRACTICE ›› apparmor

A process confined by an AppArmor profile in enforce mode tries to write outside its allowed paths. What happens?

A The operation is allowed but logged.

B The operation is blocked and logged, and the process keeps running.

C The process is killed immediately.

PROMPTS ›› apparmor

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

HINT

Start from what the service actually reads, writes and binds to, and build the profile in complain mode before you enforce it.

LOOK AT aa-genprof; the paths you worked with in Step 3.

LOG Describe your approach and the risk of enforcing too early.

HINT

Think about a legitimate action a profile did not anticipate, and where a denial would show up.

LOOK AT The system journal and /var/log/syslog for AppArmor denials.

LOG State how you verified each service still works.

2/2 hints opened

Step 7. Run Lynis again and compare

```
newadmin@ops-01:~$ sudo lynis audit system --cronjob | tee ~/lynis-after.txt
sudo: lynis: command not found
newadmin@ops-01:~$ grep -E "Hardening index|Warning|Suggestion" ~/lynis-after.txt | head -30
newadmin@ops-01:~$ grep "Hardening index" ~/lynis-before.txt
grep "Hardening index" ~/lynis-after.txt
newadmin@ops-01:~$
```

Step 7. Run Lynis again and compare

Week 3 Lab · System Hardening · Step 7 of 8

```
# Run the post-hardening audit:
sudo lynis audit system --cronjob | tee ~/lynis-after.txt

# Filter out your indices and findings from the saved logs
grep -E "Hardening index|Warning|Suggestion" ~/lynis-after.txt |

# Compare scores (Ensuring matching filenames from Step 1)
grep "Hardening index" ~/lynis-before.txt
grep "Hardening index" ~/lynis-after.txt
```

Record both scores with a brief explanation of what drove the difference. This is the most quantifiable evidence of Week 3's work.

- [x] I saved the after score and the text export (evidence items 2 and 7).
- [x] I compared the before and after files and noted what changed.

✓ Slide complete

PROMPTS ›› lynis2

Answer in your own notes first, then flip the card for a hint. Hints point you at where to look; they do not give the answer.

HINT

Match the warnings and suggestions that disappeared to the fixes you made. Did any fix change nothing, or any warning appear that was not there before?

LOOK AT `diff ~/lynis-before.txt ~/lynis-after.txt`.

LOG Record both scores and what drove the difference.

HINT

Ask whether every control suits every environment, and what each control costs to run and maintain.

LOOK AT Your remaining warnings, and what resolving each would involve.

LOG Name two remaining findings you deliberately leave, with your justification.

2/2 hints opened

Step 8. Decision log entry

Step 8. Decision log entry

Week 3 Lab · System Hardening · Step 8 of 8

Update your decision log alongside the Week 1 and Week 2 entries. They feed your final coursework report, so write them while the evidence is fresh.

Write one entry per change, using this structure and your own output:

```
## [change name]
Earlier finding it resolves (Week 1 or 2):
Fix applied (command or config):
Evidence (your own before and after output):
Why this fix:
What this does not fix:
```

You need an entry for each of these:

- Lynis score before and after (the headline evidence of Week 3)
- Services disabled or removed
- legacy-report-tool: what you fixed, in what order, and why
- File permission findings and what you changed
- Kernel parameters added
- AppArmor: which profiles are now in enforce mode
- Lynis warnings still outstanding, each with your reason for deferring it

Quality bar:

- Every claim points to your own output or to a file on the evidence checklist.
- You can say what each part of every command does.
- Outstanding warnings come with a justification: it shows you understood the finding, not that you missed it.
- Keep each entry short: your final report is 3000 words for all the weekly tasks combined.

- [x] I wrote an entry for every item above.
- [x] I took a VM snapshot and exported the VM (Checkpoint 1 complete).
- [x] I uploaded my evidence files to GitHub and committed the changes.

✓ Slide complete

Evidence checklist

Week 3 Lab · Evidence

Collect each piece of evidence, save it at the path shown, then click its status to switch it from **[pending]** to **[done]**.

#	What	Where to save	Status
1	~~Lynis hardening score - before hardening~~	`screenshots/week3-hardening/lynis-before.png`	[done]
2	~~Lynis hardening score - after hardening~~	`screenshots/week3-hardening/lynis-after.png`	[done]
3	~~`sudo ss -tulpn` after service removal - port 23 gone~~	`screenshots/week3-hardening/services-after.png`	[done]
4	~~`aa-status` showing profiles in enforce mode~~	`screenshots/week3-hardening/apparmor-enforce.png`	[done]
5	~~`stat /opt/legacy-report-tool/config.yml` showing fixed permissions~~	`screenshots/week3-hardening/config-permissions-fixed.png`	[done]
6	~~Lynis before text export~~	`configs/week3-lynis-before.txt`	[done]
7	~~Lynis after text export~~	`configs/week3-lynis-after.txt`	[done]
8	~~AppArmor status export~~	`configs/week3-apparmor-status.txt`	[done]

8/8 done
Take a VM snapshot and export after Week 3: this is Checkpoint 1 complete.

✓ Slide complete